Why Was I Rejected?

Turning credit rejections into actionable plans, and auditing who gets one

Tonantzin Real Rojas · AIPI 590: Explainable AI · Duke University

Try the app & see the code
github.com/tonantzin-real/AIPI_590_Project_1
Live demo GIF, notebook, and full results inside

1 Why this matters

In Mexico, millions of people are outside the formal financial system, and few know what lenders look for.

53% of Mexican adults have a formal financial account (vs. 73.5% in Latin America & Caribbean)
World Bank Global Findex, 2024 data

37% of adults (18–70) hold any formal credit
INEGI & CNBV, ENIF 2024

61% score below the minimum desirable level of financial knowledge
BBVA Research, 2019 (OECD/INFE method)

The problem

When an algorithm rejects someone, today's explanations (e.g. **SHAP**) say *why*: "your payment history raised your risk." That helps the lender, but not the applicant, who needs to know **what to do next**.

Audience: people applying for credit, especially first-time and underserved borrowers, not the lender.

Research questions

RQ1 How actionable are SHAP's explanations for rejected applicants?

RQ2 Can we offer realistic plans using only changes a person can make **today**?

RQ3 Is that path to approval **equally available** across sex, age, education, and marital status?

2 What I built

1. **Credit model.** LightGBM on 30,000 credit card clients (UCI, Taiwan 2005), ROC-AUC 0.78. Rejected if predicted default risk ≥ 50%.
2. **Why? (SHAP).** Which of each person's top reasons could they actually change?
3. **What to do? (DiCE).** Plans that only change what a person controls, in plain language.
4. **Fairness audit.** All 1,749 rejected test applicants, across 4 demographic dimensions.
5. **Web app** where anyone can try a profile.

Key design decision: what can a plan change?

✓ Can change
Payment status: only improve
Balance: only go down
Payment: only go up, up to the bill

🔒 Locked
Who you are: sex, age, education, marital status
The past: previous months
The lender's: credit limit

Effort = how big a plan's changes are (normalized L1 distance). Group gaps tested with Fisher's exact and Welch's t-tests, corrected for multiple comparisons.

Example: one rejected applicant

24 years old · 2-month delay · $45,000 balance · paid $500 last month

✗ Rejected: 84.5% predicted risk of default

SHAP: *why?*
1. Payment status, this month ✓ can change
2. Bank-set credit limit ✗ can't change
3. Amount paid, 2 months ago ✗ can't change

Our plan: *what to do*
1. **Catch up on payments:** 2-month delay → minimum payment made
2. **Pay more this month:** $500 → $16,281

✓ Approved: risk drops to 46.3%

3 What I found

RQ1 · SHAP explains the past, not the next step

68% of SHAP's top reasons are things the applicant **can't change today**.

49.5% of rejected applicants: the #1 reason is not actionable

27.7% get no actionable reason at all in their top 3

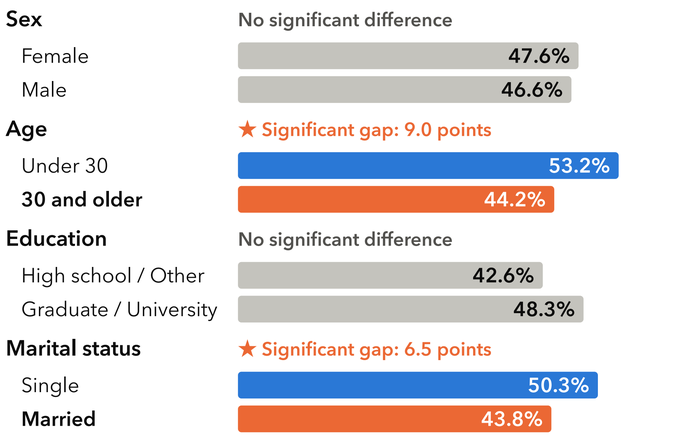
RQ2 · Realistic plans are possible for about half

47.2% of rejected applicants (825 of 1,749) get a realistic plan using only 3 things they control.

RQ3 · Equal effort, unequal access

Effort is the same for everyone, but applicants who are **30 and older** or **married** get a plan **less often**.

Out of the rejected applicants in each group, how many got a plan?



Effort, once a plan exists: about the same for every group (group ratios 0.93x to 1.01x). No difference is significant.

4 Takeaways

1. **Explanations should be actionable.** Attribution alone leaves most rejected applicants without a path forward.
2. **Audit access, not only effort.** Here the unfairness is *who gets a plan at all*.
3. **Realistic plans are possible** for about half of rejected applicants.

Limitations

- **Taiwanese data (2005)**, not Mexican; amounts are not in pesos.
- **Rejections are simulated** from predicted default risk.
- **Only 3 things can change**; no income or savings data.
- **Effort is statistical**, not the real cost for a given person.
- **Age and marital status overlap**; not tested jointly.

Risks & responsible deployment

- **Unaffordable plans** could push people to informal loans.
- **Plans are guidance**, not guarantees.
- **Explanations need audits** to avoid fairwashing.

Future work: from v1 to Mexico

- Retrain with a **Mexican institution's data**.
- **Spanish, mobile-first** app.
- Affordable, multi-month plans.
- User study with unbanked users.

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